LAB 5 EXERCISE SHEET

Q1: Write a C++ program that uses `binarySearch(data, item)` as a function. Your program should input the (sorted) array from the user and then repeatedly prompt for an item to search for. Attach code and output for best, worst, and average case.

Q2: Analyze the binary search algorithm for the best, average, and worst cases. Express your results in Big-O notation.

Q3: Input an array from the user; prompt the user again if the data entered is not sorted. Modify the binary search algorithm so that if the search is unsuccessful, the item is inserted into the array, keeping it sorted. Implement the revised algorithm in C++. Attach code and output.

Q4: Pick one real-world use of binary search and explain in two or three sentences how it maps onto the beg / mid / end idea from this lab.